

# Mohammed Alaklounk

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## EDUCATION

### Gebze Technical University

Sep 2024 – Jul 2028 (expected)

BSc Computer Engineering · GPA 3.55 / 4.00

## PROJECTS

### Scree — Falling-sand physics simulator

[Live Demo](#) · [Source](#)

*C++20, raylib, Dear ImGui, GLSL, nlohmann/json, CMake, Catch2, Emscripten/WASM, GitHub Actions*

- Built a pixel-level particle simulation on a 640×640 grid running at 60 FPS, natively and in the browser via WebAssembly, with 32 materials that flow, burn, dissolve and melt into one another.
- Optimized the update loop with 32×32 activity chunks: settled regions sleep and are skipped entirely, edits wake neighbouring chunks across borders, and only changed chunks are re-rendered.
- Designed a data-driven engine where movement, lifespans and tag-based reactions are defined in JSON and hot-reload at runtime; existing particles are remapped by name so the running scene survives a reload.
- Wrote a fault-tolerant loader with tiered error reporting (warn, reject material, reject file), an in-app material editor, and run-length-encoded canvas save/load that validates files before touching the grid.
- Implemented bloom lighting in GLSL: emissive materials render to a half-resolution buffer, pass through a separable Gaussian blur, and are composited back additively.
- Covered the parser and canvas format with 55 Catch2 test cases; GitHub Actions builds, tests and publishes Windows releases and deploys the web build to GitHub Pages.

### Latch — Logic-gate circuit simulator

[Source](#)

*C++20, raylib, Dear ImGui, CMake, Catch2*

- Built a desktop sandbox for designing and simulating digital circuits on an infinite pan-and-zoom canvas, with 7 logic gates, toggle switches and LEDs, and a one-tick propagation delay per gate.
- Modelled three-valued logic (LOW / HIGH / UNDEFINED) with per-gate lookup tables, so floating inputs are shown explicitly instead of guessed.
- Represented wires as graphs of nodes and segments with orthogonal routing; wires can branch, be tapped mid-segment, split cleanly when parts are deleted, and reroute automatically when gates move.
- Implemented snapshot-based undo/redo, rubber-band selection, copy/paste, and multiple circuit sheets sharing one component palette and clipboard.
- Isolated the simulation in a headless static library, decoupled from the UI and tested with 32 Catch2 test cases.

### The Deep Ocean — Scroll-driven web experience

[Live Demo](#) · [Source](#) · [Case Study](#)

*JavaScript, HTML, CSS, GSAP + ScrollTrigger, Lenis, Canvas 2D*

- Recreated [neal.fun's The Deep Sea](#) as a scroll from the surface to 10,924 m, placing 128 real creatures at their true depths on a CSS grid driven by a single pixels-per-metre ratio.
- Built a canvas particle system with per-particle parallax depth and a depth-driven target count (up to 300) that ramps in, tapers off and empties out by 8,000 m.
- Added a live depth meter, zone navigation, an auto-scroll tour paced in screenfuls per second and reduced-motion support, then documented the build and debugging in a written case study.

## SKILLS

**Programming:** C++ (17/20), Python, JavaScript, GLSL

**Graphics & UI:** raylib, OpenGL, Dear ImGui

**Web:** HTML, CSS, GSAP, Lenis, Canvas 2D

**Tooling:** CMake, Git, GitHub Actions, Emscripten/WASM, Catch2

**Languages:** Arabic (native), English (fluent), Turkish (B1)